Naming a Generation: Regional and Gender Patterns in 21st Century American Baby Naming

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Baby names provide a measurable way to examine cultural change, regional identity, and social trends over time. Using United States Social Security baby name records from 2000 and 2024, this study evaluated naming diversity and regional naming variation across female and male births within four United States Census regions: Northeast, Midwest, South, and West. Naming diversity was measured using a Diversity Index derived from full name distributions, while regional variation was evaluated through rank deviation between regional and national top ten name rankings. Additional statistical analyses included Kruskal-Wallis tests, pairwise Wilcoxon rank sum comparisons with Benjamini-Hochberg adjustment, and chi-square tests of regional top ten name distributions. Results showed that female names consistently exhibited greater diversity than male names and that naming diversity increased substantially between 2000 and 2024 across all regions. The South produced the highest Diversity Index values by 2024, while the West and Midwest showed the most consistent departures from national top ten rankings. Statistical testing identified significant regional differences in 2024 female rank deviation patterns, and chi-square analyses indicated significant differences in regional top ten name distributions across all year and sex comparisons. A focused analysis of the name Hudson further demonstrated that regional adoption patterns did not emerge uniformly over time, with Midwestern growth occurring earlier and more rapidly than in other regions. Overall, the findings suggest that naming practices in the United States have become increasingly diverse over time while regional identity and gender continue to shape naming behavior and the timing of popularity trends.

Introduction

Baby names provide a measurable way to examine cultural change, social identity, and regional variation over time. Names are selected within specific social contexts yet recorded consistently across decades, making them a useful dataset for studying how personal choices reflect broader cultural patterns. Naming practices respond to both long term cultural shifts and short-term social influences, making baby names a useful indicator of changing social behavior.

Prior research has shown that naming patterns are not randomly distributed across the United States. Distinct regional clusters emerge when naming frequencies are compared across states, and these relationships shift over time as cultural influences change (Barucca et al., 2015). Baby name popularity also follows highly uneven statistical distributions, with a small number of names accounting for a large share of births while many names remain rare (Li, 2012). These distributional properties support the use of concentration-based measures when evaluating naming diversity.

Short term social influences also contribute to naming variation. Specific holidays, months, and presidential birthdays have been shown to increase the likelihood of selecting related names, indicating that symbolic events can affect naming choices (Rogerson, 2016). Additional work suggests that broader naming trends are shaped more strongly by long term frequency patterns than by isolated cultural events (Ludwig et al., 2019). Annual baby name counts are recorded consistently across time, providing a useful framework for statistical analysis of naming concentration, diversity, and regional variation. Visual analyses of baby name trends have further demonstrated that changes in popularity often reflect broader social and generational shifts over time (Wattenberg, n.d.).

Using annual United States baby name records from 2000 and 2024, this study examines naming diversity and regional naming variation across female and male births in four United States Census regions: Midwest, Northeast, South, and West. Diversity is summarized using a Diversity Index derived from full name distributions, allowing comparison of how broadly births were distributed across names within each group. Regional variation was further examined using top ten rankings and mean rank deviation from national naming patterns to evaluate how naming preferences differed across regions and over time.

Objectives and Research Questions

Building on prior research examining regional and temporal naming trends, this study evaluated differences in naming diversity and top ranked naming preferences across sex, region, and time in the United States using data from 2000 and 2024. Focus was placed on measuring changes in naming concentration and identifying regional departures from national naming patterns.

Specifically, the following questions were addressed:

• Do female and male names differ in overall naming diversity?
• How does naming diversity differ between 2000 and 2024 across regions and sex?
• How do top ten name rankings shift across regions over time?
• Do regional rank deviations suggest that certain areas of the United States differ more from national naming patterns than others?

Methods

Data from 2000 and 2024 were organized by state and sex, allowing aggregation across geographic regions. Statistical analyses were conducted separately for female and male names across the four United States Census regions: Northeast, Midwest, South, and West. All reported names were retained when calculating diversity measures so that naming distributions reflected the full range of observed names rather than only the most common names. All computations and statistical procedures were completed in R (Posit Team,2025).

Naming diversity was evaluated using a Diversity Index adapted from prior work examining uneven baby name frequency distributions (Li, 2012). The index was selected because baby name usage is highly concentrated, with a relatively small number of names accounting for a large proportion of births while many names remain uncommon. Unlike simple counts of unique names, the Diversity Index incorporates the proportional distribution of births across all observed names, allowing both name variety and concentration to be considered simultaneously. Larger Diversity Index values therefore indicate that births were distributed more evenly across many names, while smaller values indicate stronger concentration among a smaller group of highly common names.

$$Diversity\ \ Index = \frac{1}{\sum_{i=1}^{k} p_i^2}$$

In the Diversity Index formula above, p_i represents the proportion of births assigned to the i^{th} name, and k represents the total number of unique names observed within the group. Squaring each proportion gives greater weight to highly frequent names, so larger sums indicate that births are concentrated among fewer names. The Diversity Index is calculated as the inverse of the summed squared proportions, so larger values reflect broader distributions of births across many names rather than concentration among a small number of dominant choices. Diversity Index values were interpreted descriptively across sex, region, and year to compare how evenly naming patterns were distributed within each group.

Regional naming differences were further evaluated using rank deviation between regional and national top ten name rankings within the same sex and year group. The top ten names were selected to focus on highly influential naming patterns while still allowing meaningful regional ordering differences to emerge. Rank deviation measured how far a regional name ranking differed from its corresponding national ranking, allowing comparison of regional departure from broader national naming preferences.

$$\text{Rank Deviation} = |\text{Rank}_{\{\text{region, i}\}} - \text{Rank}_{\{\text{national, i}\}}|$$

In the rank deviation formula above, $\text{Rank}_{\{\text{Region, i}\}}$ represents the regional rank of the i^{th} name and $\text{Rank}_{\{\text{national, i}\}}$ represents the corresponding national rank for the same name. Mean rank deviation was then calculated across the regional top ten names, where larger values indicated greater divergence from national naming patterns. Regional differences in rank deviation were evaluated using Kruskal-Wallis tests conducted separately by year and sex. When statistically significant differences were detected, pairwise Wilcoxon rank sum tests with Benjamini-Hochberg adjustment were used to identify specific regional differences. Additional chi-square tests were conducted to compare regional top ten name distributions, with Cramér’s V used as a measure of association strength.

Results

Tables 1 through 5 present national and regional top ten baby names for 2000 and 2024, with rankings generated by aggregating births within each sex and year group and ordering names by total frequency. These tables provide descriptive comparisons of how national naming patterns were reflected across regions and where regional differences in rank became visible. Table 6 summarizes regional rank deviation from national rankings, while Figures 1 and 6 illustrate changes in naming diversity and regional popularity patterns across time. Together, the results show that diversity increased between 2000 and 2024 while regional differences in rank ordering and naming distribution remained visible across the United States.

Table 1. National Top 10 Baby Names by Gender (2000 and 2024)

Rank	2000 Boy Name	2000 Girl Names	2024 Boy Name	2024 Girl Name
1.	Jacob	Emily	Liam	Olivia
2.	Michael	Hannah	Noah	Emma
3.	Matthew	Madison	Oliver	Amelia
4.	Joshua	Ashley	Theodore	Charlotte
5.	Christopher	Sarah	James	Mia
6.	Nicholas	Alexis	Henry	Sophia
7.	Andrew	Samantha	Mateo	Isabella
8.	Joseph	Jessica	Elijah	Evelyn
9.	Daniel	Elizabeth	Lucas	Ava
10.	Tyler	Taylor	William	Sofia

Table 1 establishes the national naming baseline for comparison with the regional tables. The 2000 rankings were led by Jacob and Emily, while the 2024 rankings were led by Liam and Olivia. Several names appearing in the 2000 top ten were no longer present in the 2024 rankings, showing substantial turnover in nationally preferred names across the study period. This national ordering provides a reference point for evaluating how closely regional rankings aligned with broader United States naming patterns.

Table 2. Midwestern Top 10 Baby Names by Gender (2000 and 2024)

Rank	2000 Boy Name	2000 Girl Names	2024 Boy Name	2024 Girl Name
1.	Jacob	Emily	Oliver	Charlotte
2.	Michael	Hannah	Liam	Amelia
3.	Matthew	Madison	Theodore	Olivia
4.	Nicholas	Alexis	Noah	Emma
5.	Tyler	Taylor	Henry	Evelyn
6.	Andrew	Samantha	James	Sophia
7.	Joshua	Grace	William	Eleanor
8.	Zachary	Elizabeth	Hudson	Harper
9.	Joseph	Olivia	Jack	Mia
10.	Austin	Emma	Elijah	Violet

Table 2 presents Midwestern top ten names for 2000 and 2024. In 2000, Midwestern rankings closely followed national patterns, with most differences appearing in lower ranked positions. By 2024, regional distinction became more visible through names such as Hudson, Eleanor, Harper, and Violet, which appeared within Midwestern rankings despite differing from national ordering. Compared with 2000, the 2024 rankings showed greater reshuffling across positions, particularly among male names.

Table 3. Northeastern Top 10 Baby Names by Gender (2000 and 2024)

Rank	2000 Boy Name	2000 Girl Names	2024 Boy Name	2024 Girl Name
1.	Michael	Emily	Noah	Olivia
2.	Matthew	Samantha	Liam	Emma
3.	Nicholas	Sarah	Oliver	Sophia
4.	Joseph	Hannah	Theodore	Charlotte
5.	Christopher	Ashley	Lucas	Mia

6.	Ryan	Jessica	James	Amelia
7.	Jacob	Olivia	Henry	Isabella
8.	John	Madison	Benjamin	Ava
9.	Joshua	Julia	Jack	Sofia
10.	Anthony	Brianna	Luca	Chloe

Table 3 summarizes Northeastern top ten names across both years. Northeastern rankings remained moderately aligned with national patterns while still showing several regional departures within lower ranked positions. Compared with other regions, Northeastern rankings showed less dramatic reshuffling across years, although changes in top ranked names were still visible between 2000 and 2024.

Table 4. Southern Top 10 Baby Names by Gender (2000 and 2024)

Rank	2000 Boy Name	2000 Girl Names	2024 Boy Name	2024 Girl Name
1.	Jacob	Hannah	Liam	Olivia
2.	Michael	Emily	Noah	Emma
3.	Joshua	Madison	Oliver	Amelia
4.	Christopher	Sarah	Elijah	Charlotte
5.	William	Alexis	James	Mia
6.	Matthew	Ashley	William	Sophia
7.	John	Taylor	Mateo	Isabella
8.	James	Elizabeth	Lucas	Ava
9.	Brandon	Kayla	Henry	Evelyn
10.	Nicholas	Lauren	John	Sofia

Table 4 shows that Southern top ten rankings remained closely aligned with national patterns across both years, with many names appearing in similar positions. Compared with other regions, Southern rankings showed fewer departures from national ordering and less visible reshuffling across ranks. This pattern is consistent with the relatively small rank deviation values reported later in Table 6.

Table 5. Western Top 10 Baby Names by Gender (2000 and 2024)

Rank	2000 Boy Name	2000 Girl Names	2024 Boy Name	2024 Girl Name
1.	Jacob	Emily	Liam	Olivia
2.	Daniel	Ashley	Noah	Emma
3.	Michael	Samantha	Mateo	Mia
4.	Joshua	Jessica	Oliver	Sophia
5.	Andrew	Hannah	Santiago	Amelia
6.	Anthony	Alexis	Sebastian	Camila
7.	Matthew	Madison	Theodore	Isabella
8.	Jose	Alyssa	Julian	Charlotte
9.	Christopher	Sarah	Elijah	Sofia
10.	David	Elizabeth	Ezra	Evelyn

Table 5 shows the strongest regional distinction by 2024, with several names appearing in Western rankings that did not rank nationally in the same positions. Names such as Mateo, Santiago, Sebastian, Julian, Ezra, Camila, and Sofia contributed to greater regional variation alongside nationally common names. Compared with other regions, the Western rankings reflected the broadest departures from national ordering across both years.

Table 6: Regional Differences in Top 10 Baby Name Rank Deviation by Gender and Year(2000-2024)**

Region	2000 Female	2000 Male	2024 Female	2024 Male
Midwest	3.5	3.2	3.1	3.3
Northeast	5.1	4.0	1.8	3.0
South	1.5	3.5	0.2	2.3
West	2.6	6.1	1.8	6.6

Table 6 summarizes mean rank deviation, or the average distance between each region's top ten names and the corresponding national top ten rankings. Larger values indicate that a region's most common names appear farther from their national positions, while smaller values indicate closer alignment with national ordering. The West showed the largest male rank deviation in both 2000 and 2024, increasing from 6.1 to 6.6, while the South showed the smallest deviations overall, particularly for 2024 female names.

Kruskal-Wallis tests were used to compare regional rank deviation within each year and sex group. Statistically significant regional differences were detected only for female names in 2024, $\chi^2(3) = 13.37$, $p = 0.004$. Pairwise Wilcoxon tests with Benjamini-Hochberg adjustment showed significant differences between the South and Midwest as well as between the West and South. No statistically significant regional differences were detected for female names in 2000, male names in 2000, or male names in 2024. Additional chi-square tests comparing regional top ten name distributions were statistically significant across all year and sex groups after Benjamini-Hochberg adjustment. Cramér's V values ranged from 0.37 to 0.56, indicating moderate regional differences in the distribution of top ranked names across regions. Full statistical output is provided in Appendices A through D.

Figure 1: Regional Patterns in Baby Name Diversity Index Across Gender and Year

	Female		Male	
Region	2000	2024	2000	2024
Midwest	214.4	389.3	158.5	328.0
Northeast	224.4	390.0	133.7	302.2
South	257.6	456.8	178.4	353.8
West	311.6	403.9	194.4	320.6

Figure 1 shows that naming diversity increased across every region between 2000 and 2024. Higher Diversity Index values indicate that births were distributed across a wider variety of names, while lower values indicate greater concentration among fewer highly common names. Female diversity values remained consistently higher than male values across all regional comparisons, indicating broader distributions of female names throughout the study period. In 2000, the West showed the highest diversity values for both sexes, while by 2024 the South showed the highest overall diversity values, especially among female births. Southern female diversity reached 456.8, substantially exceeding all other regional comparisons shown in the figure. Male diversity values also increased sharply across every region between 2000 and 2024, although female diversity remained consistently higher overall.

Regional separation became more visible by 2024, particularly among female names. This pattern aligns with the statistically significant regional differences detected for 2024 female rank deviation in Table 6. Although the South remained relatively close to national top ten ordering, the broader Diversity Index values suggest that regional variation extended beyond only the highest ranked names. Together, these results indicate that regional naming differences became increasingly visible when full naming distributions were considered rather than only top ranked positions.

The broad regional differences shown in Figure 1 suggest that naming trends did not develop identically across the United States. Figure 2 narrows the focus to a single rapidly rising name to illustrate how regional popularity patterns changed across time.

Hudson was selected because it rose sharply in national popularity during the study period, reaching rank 22 nationally by 2024, while also reaching rank 8 in the Midwest and showing noticeably earlier and stronger regional adoption there compared with other parts of the United States. The large difference between the national and Midwestern rankings made Hudson a useful example of how regional naming trends can separate from broader national popularity patterns even when a name experiences substantial growth overall.

Figure 2: Regional Growth in the Popularity of Hudson, 2000 to 2024

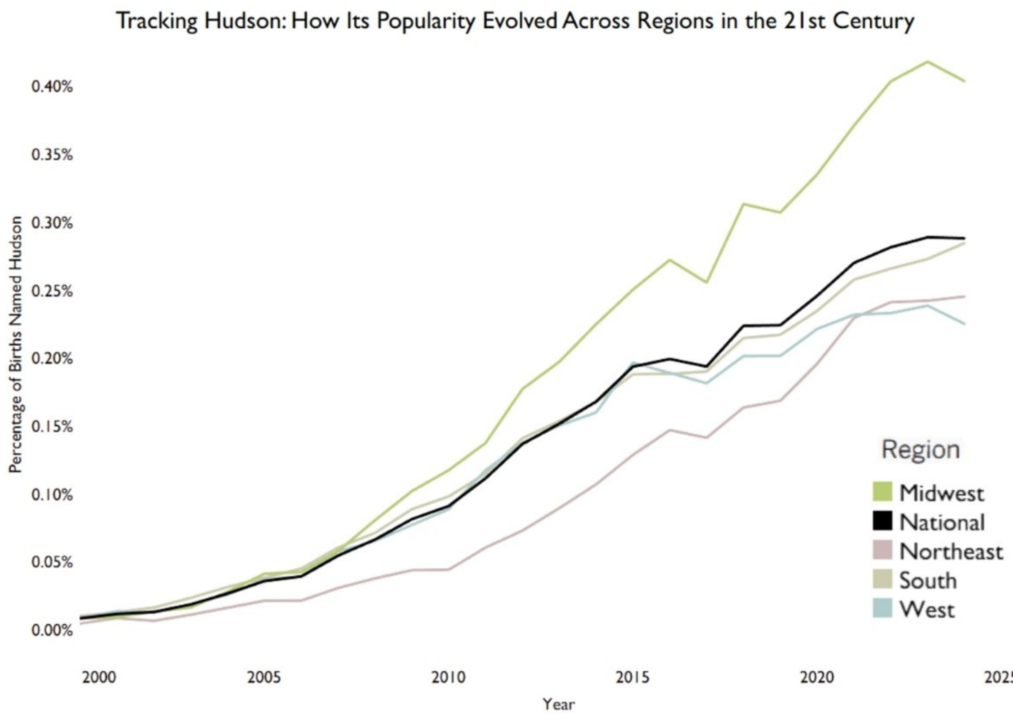

Figure 2 traces the rise of Hudson across regions between 2000 and 2024. Although all regions followed a general national upward trend, the timing and magnitude of growth differed substantially across regions. The Midwest separated earliest and remained above all other regions after the early 2010s, eventually exceeding 0.40% of births by the end of the study period. In contrast, the Northeast increased more gradually and remained below national levels for much of the timeline. The South moved closer to national levels later in the study period, while the West remained relatively close to the national trajectory overall.

The widening separation between the Midwest and other regions suggests that rapidly rising names may diffuse unevenly across the United States rather than increasing at the same pace everywhere simultaneously. While Hudson became nationally popular over time, the figure demonstrates that regional adoption timing and intensity still varied substantially even under a shared national upward trend.

Discussion & Next Steps

Baby naming patterns varied across gender, region, and time, although changes did not occur uniformly across all dimensions. Female names remained more diverse than male names across every regional comparison, while diversity increased substantially between 2000 and 2024 for both sexes. This pattern suggests continued movement away from highly concentrated distributions and toward broader variation in commonly used names over time.

Regional differences remained visible even as diversity increased nationally. The South produced the highest Diversity Index values by 2024, especially among female births, indicating that births were distributed across a wider range of names than in other regions. In contrast, the West showed the largest departures from national top ten ordering, particularly among male names, where mean rank deviation exceeded all other regional comparisons in both years examined. These findings suggest that regional naming identity may appear differently depending on whether attention is focused on the most common names or on the broader naming distribution overall.

Statistical testing further supported this distinction. Kruskal-Wallis testing identified statistically significant regional differences only for female rank deviation in 2024, while chi-square tests detected significant differences in regional top ten name distributions across every year and sex group examined. Together, these findings suggest that some regional naming differences become more visible when evaluating overall name distributions rather than only average shifts in rank positions.

The contrast between diversity measures and rank deviation also highlights that regional naming patterns cannot be summarized using a single metric alone. A region may maintain nationally familiar top ranked names while still exhibiting substantial variation beyond the highest ranks, as seen most clearly in the South. Conversely, the West combined broad diversity with repeated departures from national ordering, indicating stronger regional separation in both common and less frequent naming choices.

Temporal change remained the strongest overall pattern observed in the study. Across all regions, diversity increased substantially between 2000 and 2024, suggesting continued expansion in naming variety over time. At the same time, Figure 2 demonstrated that regional timing in name adoption remained uneven, with Hudson separating earlier and more strongly in the Midwest despite broader national growth. The continued national rise of Hudson beyond the study period, reaching rank 17 nationally in 2025, further suggests that regional adoption patterns identified in this study may continue evolving after names achieve broader national popularity.

Several extensions could strengthen future analysis. Dividing regions into smaller census divisions or individual states may reveal local variation that broader regional groupings do not fully capture. Additional comparisons using naming records from countries such as the United Kingdom, New Zealand, or Australia could help determine whether similar regional and temporal naming patterns appear internationally. Future work could also examine whether linguistic structure, phonetic patterns, spelling variation, or names associated with different cultural and language backgrounds contribute to regional naming differences across time.

References

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Appendix

Appendix A. Kruskal-Wallis Test Results

Kruskal-Wallis tests were conducted to evaluate regional differences in mean rank deviation within each year and sex group.

Year	Sex	Statistic	p-value	Parameter
2000	F	4.08	0.25	3
2000	M	1.45	0.69	3
2024	F	13.37	0.004	3
2024	M	2.40	0.49	3

Appendix B. Pairwise Wilcoxon Rank Sum Comparisons

Pairwise Wilcoxon tests with Benjamini-Hochberg adjustment identified statistically significant regional differences only among selected 2024 female comparisons.

Year	Sex	Region A	Region B	p-value
2024	F	South	Midwest	0.004
2024	F	South	West	0.033

Appendix C. Chi-Square Test Results for Regional Name Distributions

Chi-Square tests identified statistically significant regional differences in top ten name distributions across all year and sex comparisons. Cramér's V values are reported as measures of association strength.

Year	Sex	Statistic	df	Adjusted p-value	Cramér's V
2000	F	132,127.2	51	0	0.483
2000	M	253,562.8	57	0	0.555
2024	F	48,644.6	42	0	0.373
2024	M	105,073.1	54	0	0.498

Appendix D. Selected R Code for Rank Deviation and Statistical Testing

```
# Regional and national rankings

regional_ranks <- capstone_clean %>%

  group_by(region, year, sex, name) %>%

  summarise(total_births_region = sum(count), .groups = "drop") %>%
```

```
  group_by(region, year, sex) %>%
  mutate(rank_region = rank(-total_births_region, ties.method = "first")) %>%
  ungroup()

national_ranks <- capstone_clean %>%
  group_by(year, sex, name) %>%
  summarise(total_births_nat = sum(count), .groups = "drop") %>%
  group_by(year, sex) %>%
  mutate(rank_nat = rank(-total_births_nat, ties.method = "first")) %>%
  ungroup()

# Rank deviation for regional top ten names
rank_dev_top10 <- regional_ranks %>%
  left_join(national_ranks, by = c("year", "sex", "name")) %>%
  mutate(rank_diff = abs(rank_region - rank_nat)) %>%
  filter(rank_region <= 10)

# Kruskal-Wallis tests
kruskal_rank_tests <- rank_dev_top10 %>%
  group_by(year, sex) %>%
  summarise(
    test = list(kruskal.test(rank_diff ~ region)),
    .groups = "drop"
  ) %>%
  mutate(tidy = map(test, broom::tidy)) %>%
  unnest(tidy)

# Pairwise Wilcoxon tests
pairwise_rank_tests <- rank_dev_top10 %>%
  group_by(year, sex) %>%
  group_modify(~ {
```

```
  pw <- pairwise.wilcox.test(
    x = .x$rank_diff,
    g = .x$region,
    p.adjust.method = "BH",
    exact = FALSE
  )

  as.data.frame(as.table(pw$p.value)) %>%
    filter(!is.na(Freq)) %>%
    rename(region_1 = Var1, region_2 = Var2, p_adj = Freq)
}) %>%
ungroup()
```